Deep Learning Enhanced Virtual Sensing for Water Quality Estimation

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Efficient monitoring of water quality parameters is crucial for the successful management of aquaculture systems, ensuring the health and growth of aquatic life. Traditional sensor-based systems are often cost-prohibitive, limiting their adoption, particularly in resource-constrained settings. In this research, a cost-effective method that leverages virtual sensing to estimate dissolved oxygen (DO) levels using solubility and interdependency equations is introduced. Furthermore, a novel approach to Water Quality Monitoring (WQM) in aquaculture, utilizing a Virtual Sensing (VS) system integrated with Deep Learning (DL) is proposed to accurately estimate DO. To the best of the authors' knowledge, this research marks the first proposal of integrating Virtual Sensing and Deep Learning to reduce the cost of sensor-based WQM systems for aquaculture.

Virtual Sensing Using Solubility Equations

Virtual sensing (VS) methodology leverages established solubility equations and inter-dependency relations to estimate water quality parameters, with a specific focus on dissolved oxygen (DO). By utilizing mathematical models derived from comprehensive datasets, the VS system offers a cost-effective and practical alternative to physical sensors. The flexibility of this approach allows to estimate DO concentrations and Salinity irrespective of the source of water, making it applicable to various water bodies.

VS Capabilities Irrespective of Water Source

The VS system, based on solubility equations, is designed to estimate water quality parameters across diverse water sources. Through mathematical models, the system predicts parameters that are challenging to measure directly through physical sensors. This flexibility ensures accurate estimations for a wide range of scenarios.

Deep Learning Enhancement for Accuracy

To address the Controller's question regarding the learning mechanism and real-time error correction, the VS system incorporates deep learning recurrent neural network (DL-RNN) models with Virtual Sensing mechanism. This mechanism aims to compensate for potential inaccuracies in the VS-based DO estimation, particularly when dealing with new, untrained water bodies in real time.

DL-RNN Training and Error Prediction

The DL-RNN model undergoes comprehensive training on various water quality datasets to predict errors in VS-based DO estimates. This rigorous training process ensures the adaptability

and high predictive accuracy of the model across diverse water sources. The utilization of multiple datasets contributes to the robustness of the DL-RNN, allowing it to capture the intricacies and variations present in different water quality scenarios.

Real-Time Measurements and DL Enhancement

In real-time scenarios, salinity and temperature are measured, and the VS system estimates DO using solubility equations. The DL-RNN model is engineered to predict errors in virtual sensing without requiring an exhaustive study of specific dataset characteristics. This unique attribute enables the model to generalize its learning, making it exceptionally versatile across a spectrum of water quality scenarios. By focusing on error prediction, rather than being dependent on a specific dataset, the DL-RNN ensures adaptability to new water bodies in real time, enhancing the system's overall effectiveness. The DL-RNN model predicts errors based on the dynamic characteristics of the specific water source. By subtracting the predicted error from the VS-estimated DO, enhanced DO estimate that is more accurate and reflective of real-time conditions is obtained.

Conclusion

In conclusion, the VS system, leveraging solubility equations and DL-RNN models, demonstrates the capability to estimate water quality parameters irrespective of the source of water. The integration of DL-RNN models through the DLEVS mechanism enhances the accuracy of DO estimation by predicting and compensating for potential errors. Real-time measurements of salinity and temperature, coupled with virtual sensing and DL enhancement, provide a comprehensive and accurate approach to water quality estimation. To the best of the authors' knowledge, this research marks the first proposal of integrating Virtual Sensing and Deep Learning to reduce the cost of sensor-based WQM systems for aquaculture. The findings contribute to advancing sustainable aquaculture practices by providing a more accessible and accurate method for monitoring water quality parameters.